

Causality Assessment

Task ID: task3

Topic: Evaluation of the relationship between a drug and an adverse drug reaction

1. Introduction

Causality assessment is the process of evaluating whether a suspected medicine is responsible for an adverse drug reaction (ADR). It is performed by reviewing the patient's clinical history, the medicine taken, the timing of symptoms, improvement after stopping the drug, and other possible causes. The result helps classify the reaction as Certain, Probable, or Possible.

2. Aim

To evaluate the relationship between a suspected drug and an adverse reaction using basic causality assessment principles.

3. Objectives

- To understand the concept of causality assessment.
- To study the relationship between drug administration and symptom onset.
- To assess dechallenge and rechallenge information.
- To consider alternative causes of the reaction.
- To classify the suspected ADR as Certain, Probable, or Possible.

4. Parameters Used for Causality Assessment

Parameter	Explanation
Temporal relationship	The reaction occurs after the suspected medicine is started.
Known reaction	The symptoms are consistent with a known adverse effect of the drug.
Dechallenge	The symptoms improve after the suspected medicine is stopped.
Rechallenge	The reaction returns after restarting the medicine, if this occurs naturally or is clinically justified.
Alternative causes	Other medicines, diseases, infections, food, or environmental factors are considered.
Clinical evidence	Examination findings, laboratory reports, or other evidence support the reaction.

5. Case Study

A 25-year-old male patient was prescribed Amoxicillin 500 mg orally for a bacterial throat infection. After taking the medicine for approximately two days, he developed red skin rashes and itching over the arms and chest. He had not experienced similar symptoms before starting the medicine. After medical advice, Amoxicillin was stopped and the symptoms gradually improved. Rechallenge was not performed.

6. Case Analysis

Assessment point	Observation
Suspected drug	Amoxicillin 500 mg

Assessment point	Observation
Adverse reaction	Red skin rash and itching
Time of onset	Approximately two days after starting treatment
Dechallenge	Symptoms gradually improved after stopping the suspected drug
Rechallenge	Not performed
Alternative causes	No other cause is identified in this educational case
Final classification	Probable

7. Classification of Causality

Category	Meaning
Certain	A clear time relationship exists, other causes are unlikely, and strong confirming evidence is available
Probable	A reasonable time relationship exists, other causes are unlikely, and the reaction improves after withdrawal
Possible	A reasonable time relationship exists, but another disease or medicine could also explain the reaction

8. Interpretation

In this case, the symptoms appeared after Amoxicillin was started and improved after the medicine was withdrawn. The symptoms are compatible with a possible allergic skin reaction. Since other causes are not identified in this educational case, the relationship is classified as Probable. It is not classified as Certain because there is no rechallenge or strong confirmatory evidence.

9. Conclusion

Causality assessment helps determine how strongly a medicine is associated with an adverse reaction. The main points are the time relationship, known drug effects, dechallenge, rechallenge when available, and exclusion of alternative causes. In the present illustrative case, Amoxicillin is the probable suspected drug.

10. Note

This report is prepared for educational submission based on the Task 3 description. The case is illustrative and is not a real patient record. It should not be used for diagnosis or treatment decisions.

11. References

- World Health Organization (WHO): Pharmacovigilance and adverse drug reaction monitoring.
- Standard textbooks of Pharmacology and Clinical Pharmacy.
- General principles of adverse drug reaction causality assessment and reporting.